

Owner's Manual, Installation Instruction and Trouble Shooting Guideline:

Applies to the following models:

1. Toyota Tundra (2014 – 2021)

Toyota Tundra (2014 – 2021) Tesla Series Wiring Harness Introduction:



Installation Video Instruction (Toyota Tundra | 2014-2021):



[Protocol Car Model/AC Settings]

Access the CarModel / AC Settings:

- Go to: **Settings** → **System** → **Setting Guide** → **Car Model / AC**
- Enter password: **666888**
- Ensure the **Protocol** is set to **BNR with AC**.

Verify that the **Car Model** is selected correctly.

- Automatic Climate Control: **Toyota – Tundra HI**
- Manual Climate Control: **Toyota – Tundra LO**
- Automatic Climate Control with Heated Seat: **Toyota – Tundra HI w/seat heaters**

15:51

Protocol▼
Import table
Restore

Current: Tundra HI
Protocol: BNR with AC

Toyota	Tundra LO
Honda	Tundra HI
Nissan	Tundra Hi w/seat heaters
Jeep	
Dodge	
LAND ROVER	
Maserati	

26.0°C

 10

26.0°C

[Wiring Instruction]

Note: There are **TWO** different power harnesses included in the package. One for 2014–2019 models and another for 2020–2021 models. You will only need to use **ONE** of the harness that matches your vehicle.

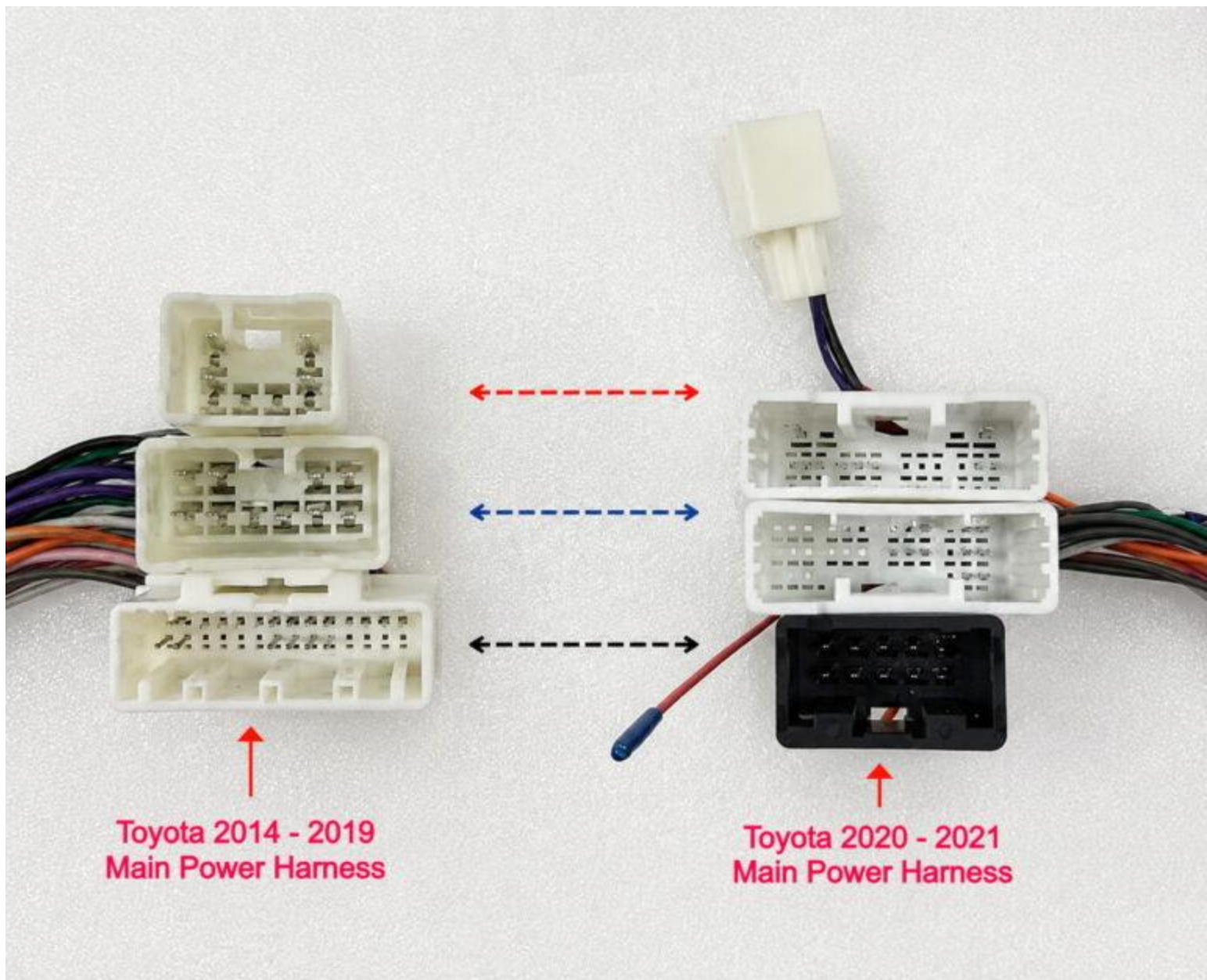
If the correct harness for your vehicle is missing from the package, please contact us and we will be happy to assist you.

**2014 - 2019
Main Harness**



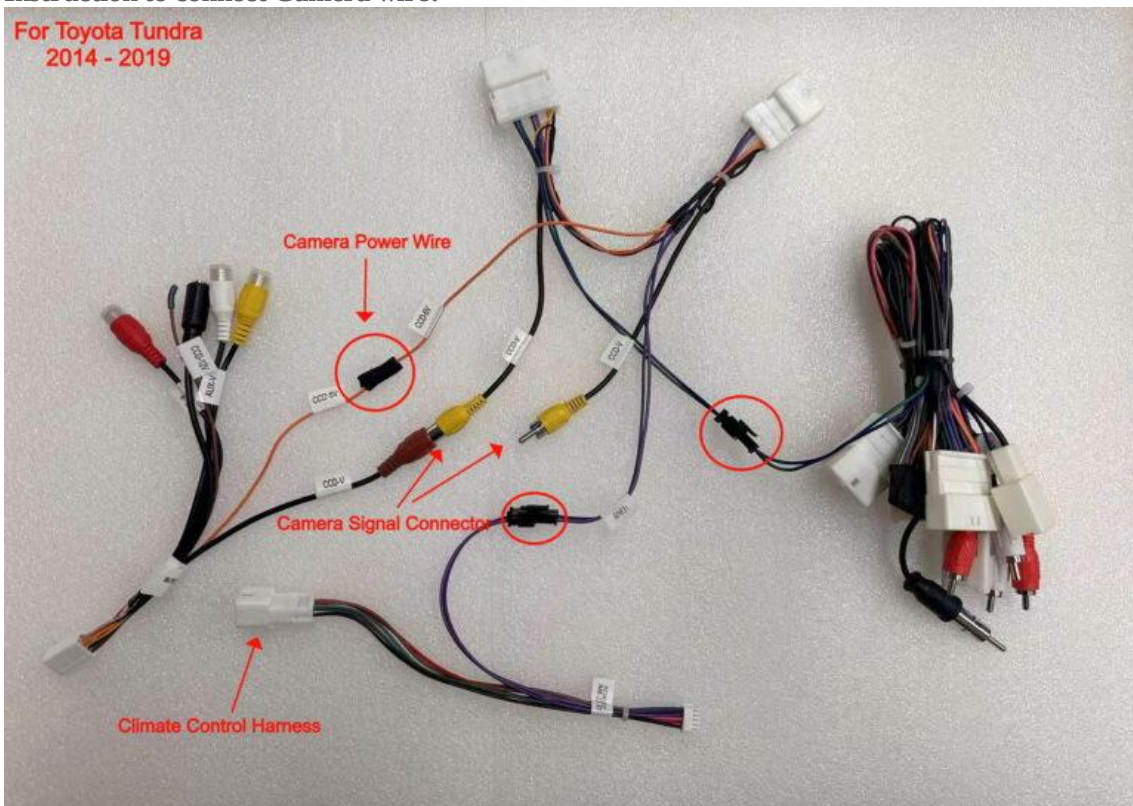
**2020 - 2021
Main Harness**



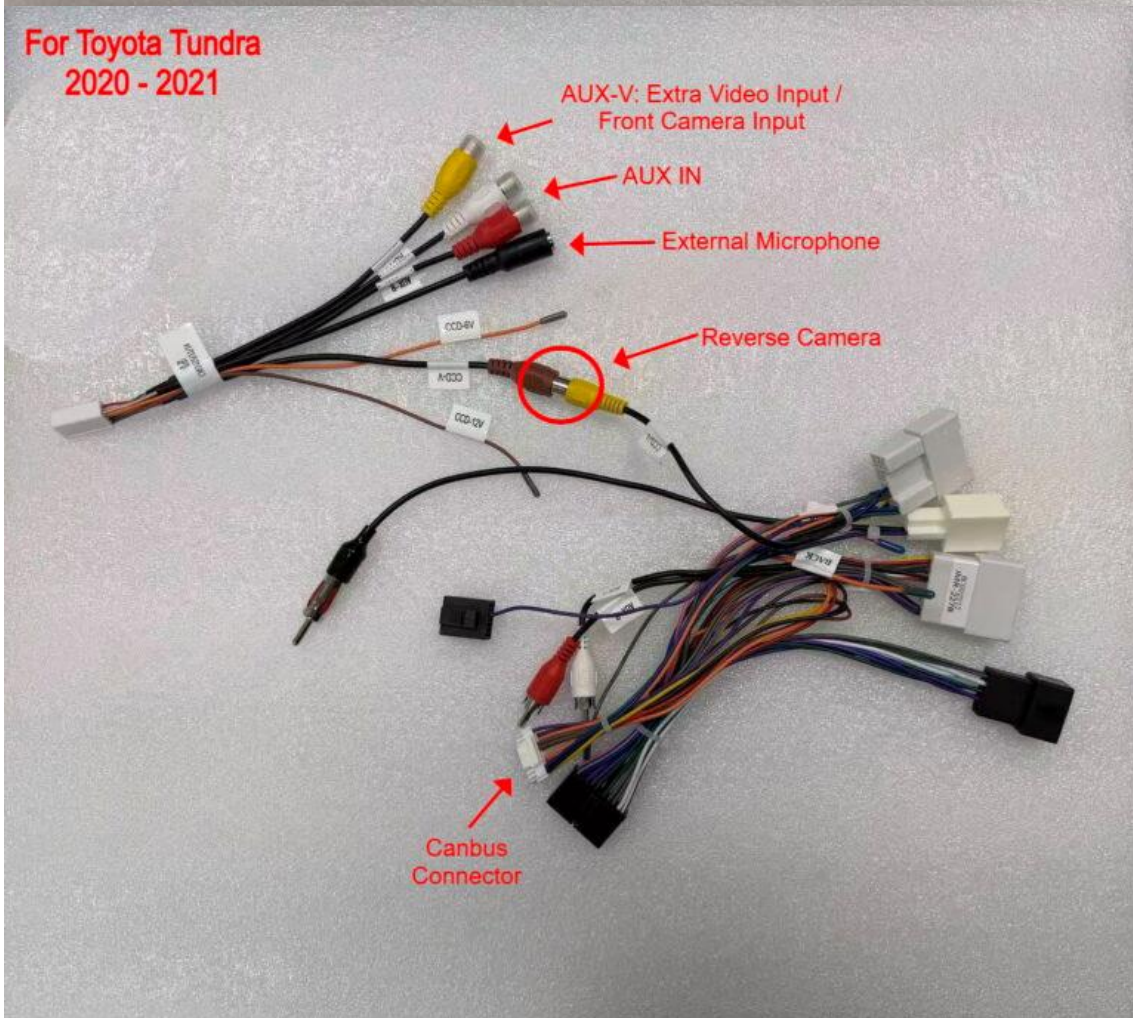


Instruction to connect Camera wire:

For Toyota Tundra
2014 - 2019

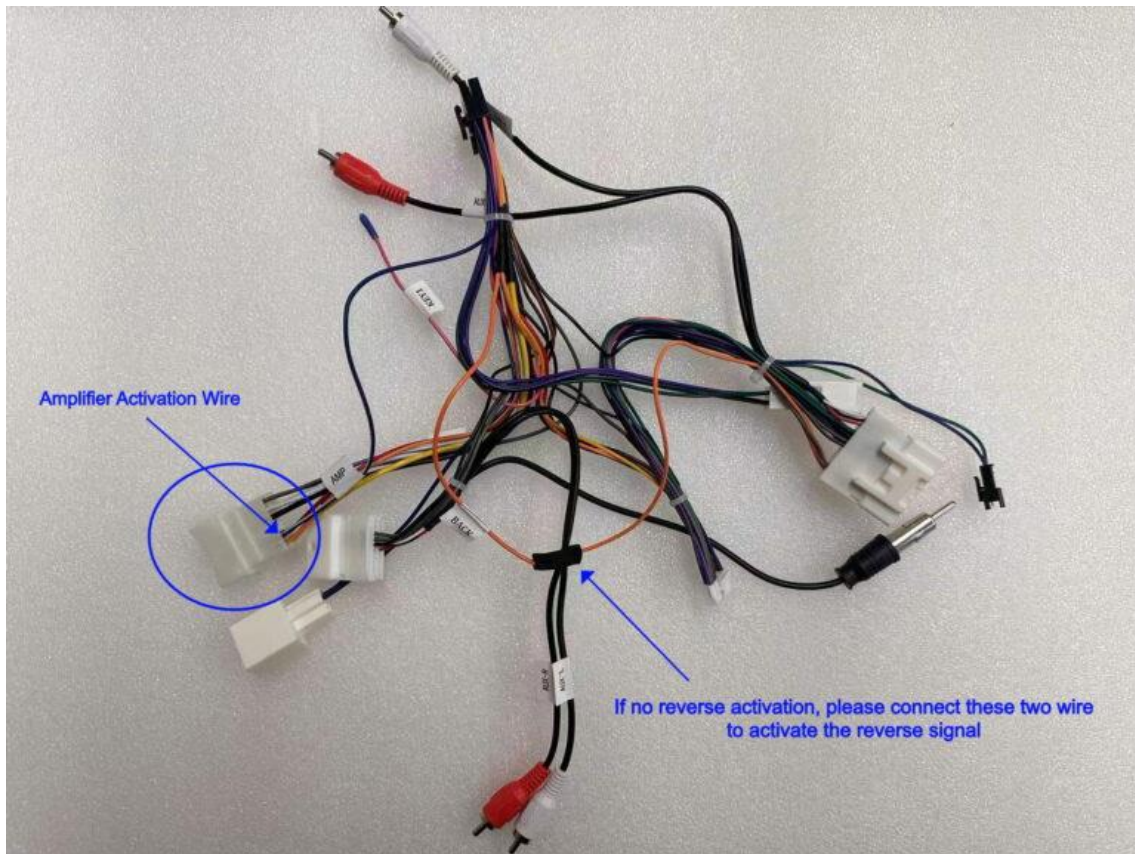


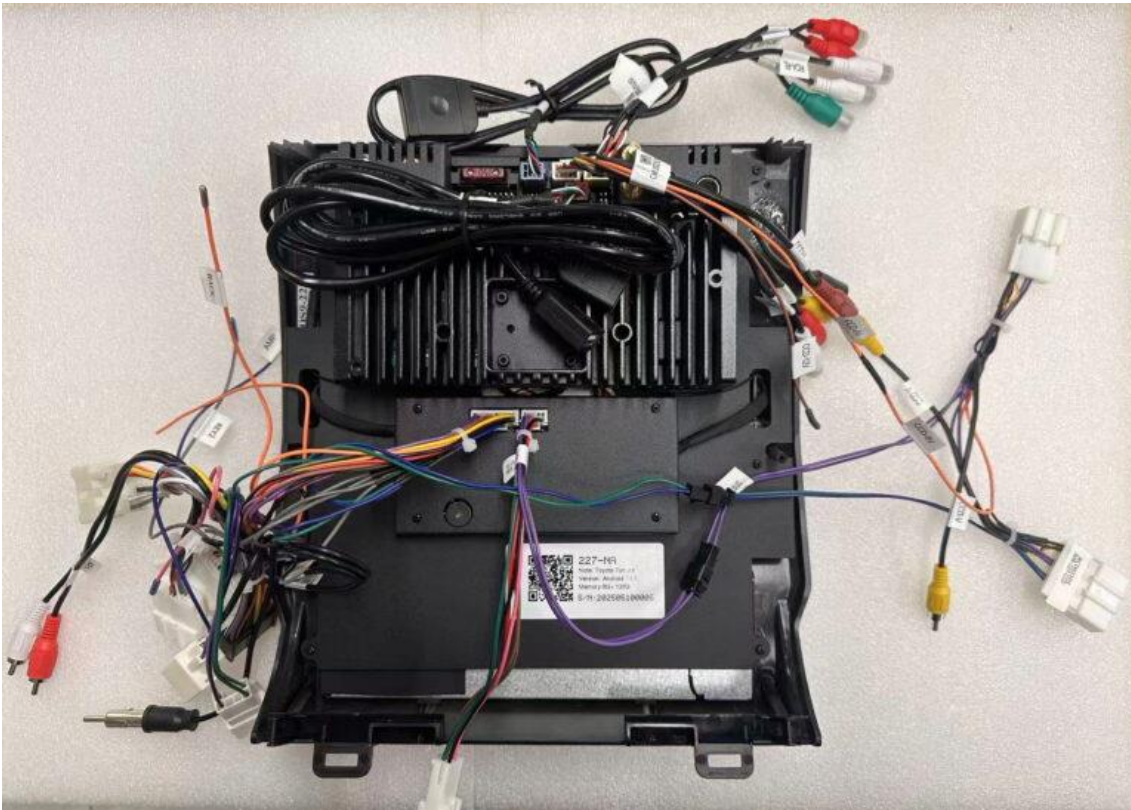
For Toyota Tundra
2020 - 2021



After installation, if the reverse camera is not triggered automatically when shifting into reverse, please connect the reverse activation wire as shown in the diagram below. Otherwise, do not connect this wire.

Additionally, if there is no audio from the speakers after installation, you may need to connect the blue AMP wire to the corresponding pin on the original harness by splicing it accordingly.





[Original Backup Camera]

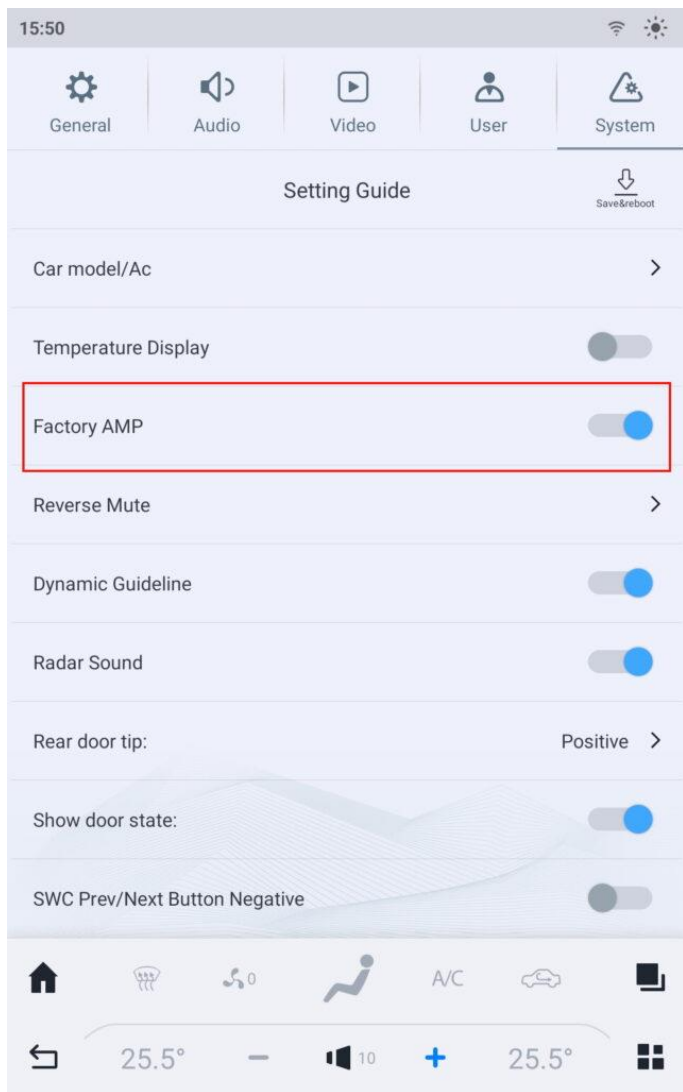
If you have issues with the original camera after the installation, please refer to the video below:



[Audio Trouble Shooting] – For JBL Vehicle

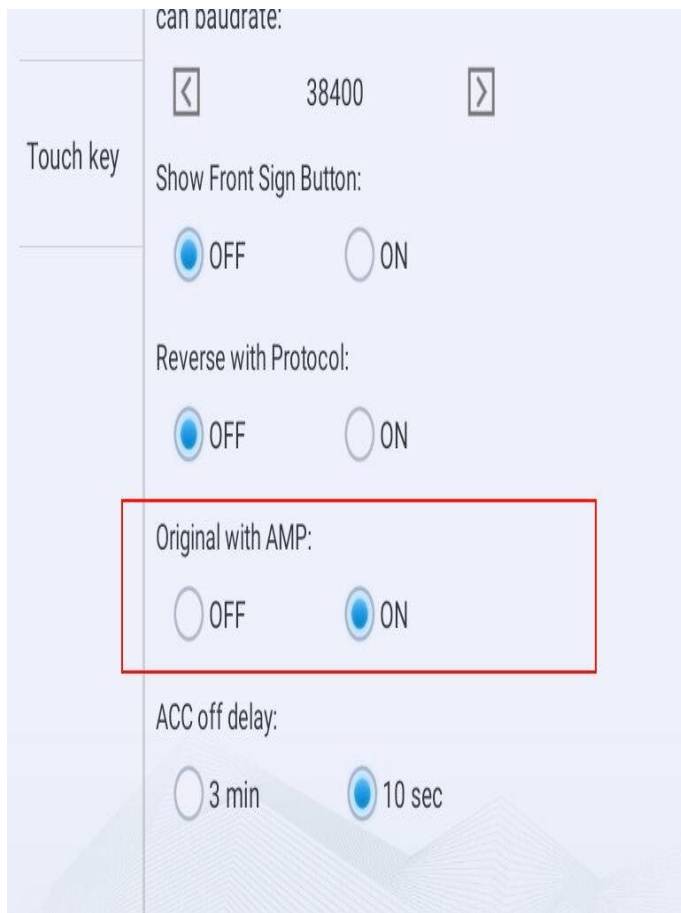
1. After the installation, if your speakers are not fully functional. You can follow the steps below:

- Go to Settings – System – Setting Guide
- Turn on “**Factory AMP**”
- Select “Save & Reboot”



2. Go to Setting – System – Factory Setting (Password: 7890)

- Turn on **“Original with AMP”**
- Select **“Save & Reboot”**



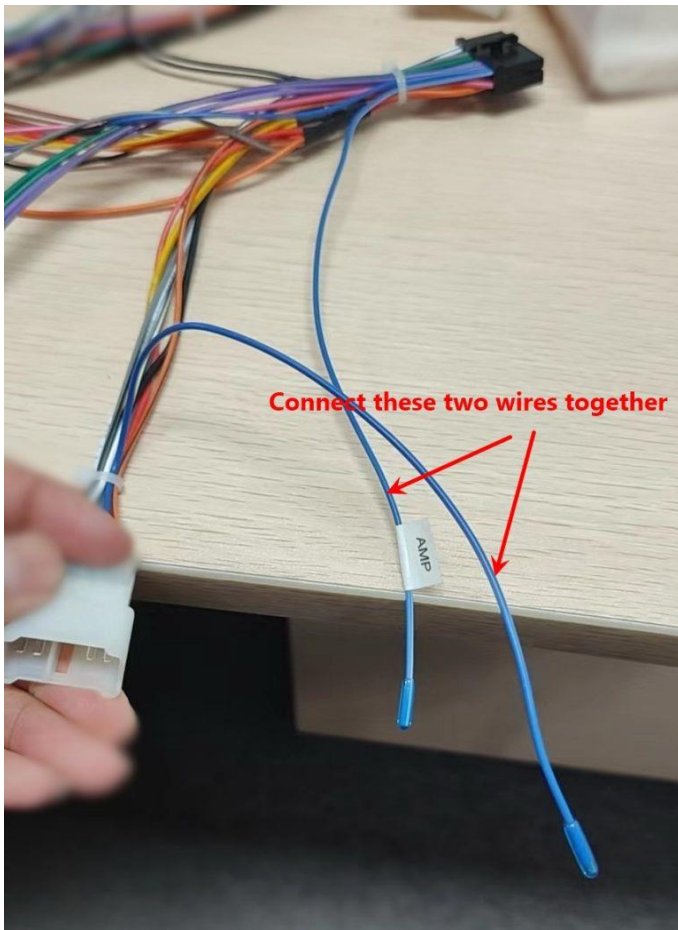
3. Go to Car Setting – Amp Info

- Increase the value of “**Cur Vol**” up t the range between 35-40.

Middle:	—	0	+
Cur Vol:	—	11	+
AMP Switch:	—	Power Off	+
Surround:	—	OFF	+
Mute Switch:	—	OFF	+
Beep:	—	OFF	+

[Audio Issues]

If your car comes with original amplifier and has no sound after the installation, please connect the wires shown as below to activate the original amplifier:



[How to use / setup the new infotainment system – Comprehensive Guideline]

The video tutorial below will show you step by step how to use and setup the infotainment system at the beginning. Please feel free to watch this video if you have any questions on how to operate the system.



How to change wallpaper on your tesla infotainment system?

1. **Switch to Modern Theme**
Go to **Settings** → **General** → **Theme**, and select **Modern**.
(Wallpaper is only supported in Modern theme.)
2. **Install Google Wallpaper**
Open the **App Store**, download **Google Wallpaper**, and install it.
3. **Apply Wallpaper**
Open the **Google Wallpaper** app and choose your new wallpaper.



How to change the Weather Location on the main interface:



If you have any question about how to setup or use the system after the installation, please refer to the video below:

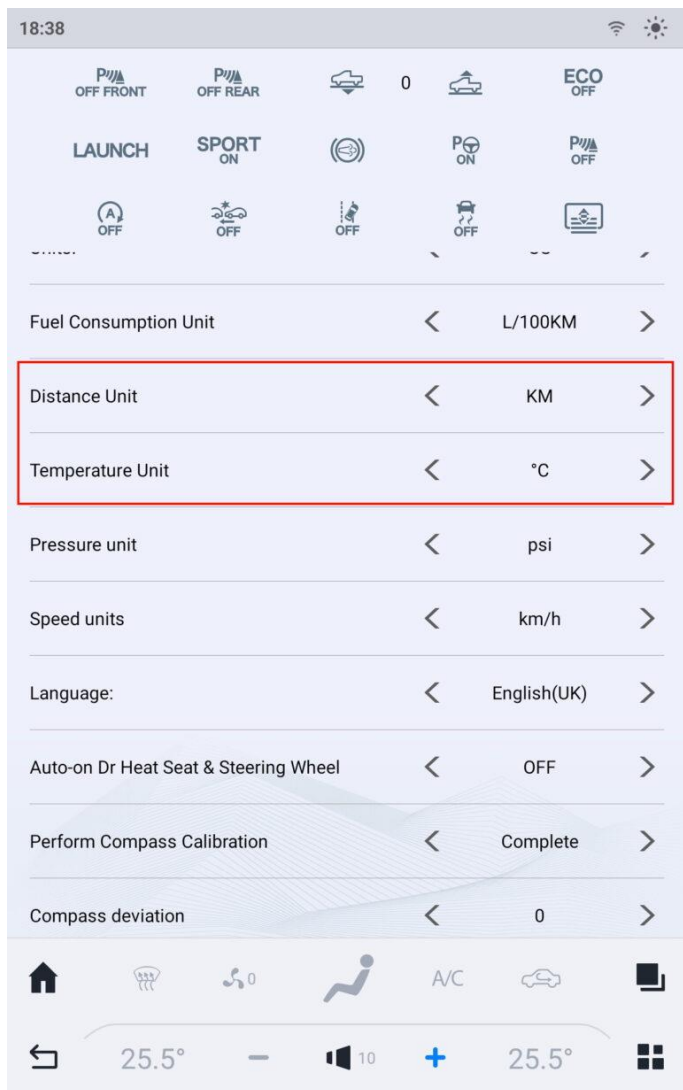


[How to change Temperature units (°F / °C)]

In order to change the temperature unit, you can simply follow the steps below to change the temperature unit:

1. Go to Console / Car Setting
2. Adjust the Temperature unit from the menu





[Aftermarket Camera Installation Instruction (Front Camera / Reverse Camera)]

If you are trying to install an aftermarket Front Camera or Reverse Camera, please feel free to take a look here: [HERE](#)

[Bakup Camera / Reverse Guideline]

You can turn ON/OFF the reverse guideline from the system by following the steps below:

Setting – System – Setting Guide – Dynamic Guideline

[BakupCamera / Reverse Mirrored Image]

If you have mirrored image when reverse (Left side become Right side / Right side become Left side), you can follow the steps below:

1. Change the gear to reverse and bring up the reverse screen first
2. Click on the Reverse setting menu on the bottom left corner
3. Adjust the settings of the option “Mirror”

[Reset Procedure]

Hardware Reset:

If the screen stay blank / black when you turn on the vehicle, you can use a small pin (such as Sim Card Tray Eject Pin) to click on the reset button. The reset button is usually located at the front of the unit and labelled as “RST”. Once you press the RESET button, the system will restart.

Note: This will not erase the softwares that you installed on the unit recently, also, it will not change the settings back to factory default state.

Please note: Do not use anything larger than the ejector pin, as it could damage or crack the screen.

Software Factory Reset:

If you want to reset the software and settings of the system back to factory state, you can follow the steps below:

1. Go to Setting – System
2. Scroll down the menu and select “Factory Reset”
3. Erase All Data

After a factory reset, if the CarAuto application (CarPlay/Android Auto) asks for activation, please follow these steps:

1. Connect to Wi-Fi and ensure the internet is working properly.
2. Verify that the time zone and system time are correctly set for your location.
3. Open the Chrome browser and try to play a video on YouTube to confirm that the internet connection is working properly.
4. Close the CarAuto app completely.
5. Reopen the CarAuto app.

The Apple CarPlay / Android Auto should activate automatically as long as the internet connection is active.

Note: This will erase the software that you installed on the unit recently, it will also reset the unit back to factory condition. If the reverse camera is not working after the reset, you can follow the steps below:

1. Go to settings – Systems – Factory Setting (Password: 7890) – Reversing Format – Select “CVBS”

[Screen Calibration]

If the screen has some touch issue, you can re-calibrate the screen by following the steps below:

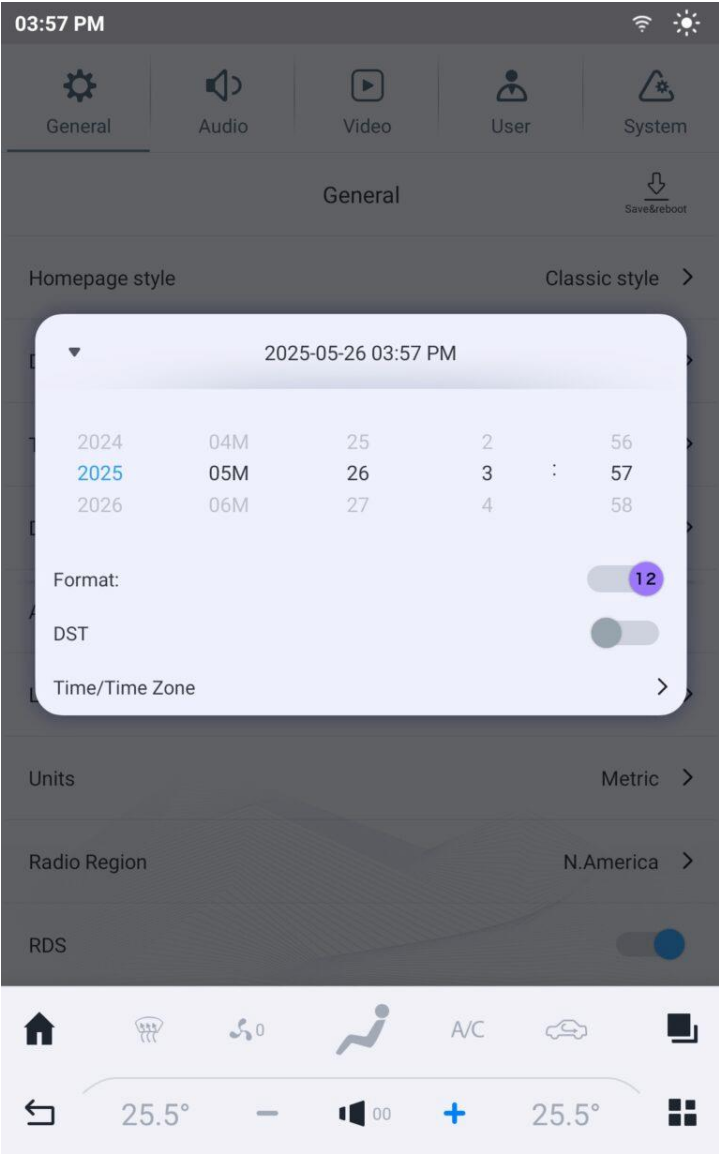
1. Hold FIVE fingers anywhere on the screen for around 10 seconds.
2. Follow the instruction on the screen
3. Click Save

[WiFi Connectivity]

In order to connect to WiFi, first you will need to make sure you are not using wireless Apple CarPlay or Android Auto at the same time. If you have connected to wireless Apple CarPlay, please turn it off from the cellphone first. For more details, please refer to the section below [Apple CarPlay].

1. Pull down the menu from the top edge of screen, hold the WiFi icon to open the WiFi connection menu.
2. Connect the WiFi to the hotspot of the cellphone.
3. Go to Setting – System – Android – Time Setting
4. Turn ON Network Time
5. Turn OFF auto time zone and manually adjust the time zone according to your area
6. Make sure the time on the upper left corner of the screen matches with your area's local time.
7. Once the time matches with the local time, WiFi should work.
8. To check whether the internet connection is working, you can simply open YouTube in the internet browser. If it loads, the connection is successful.

Secondly, you can go to Setting from the main Android interface, select Time and manually change the date and time under this menu.



[Weather]

How to change the Weather Location on the main interface:



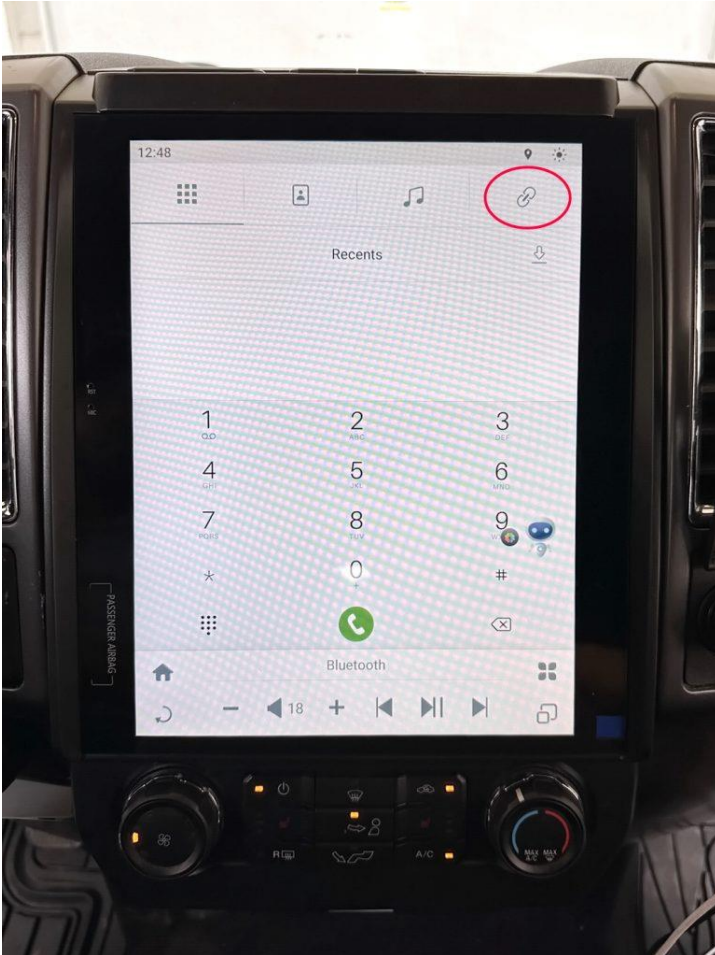
How to change the Weather Location on the main interface:



[Bluetooth]

If you are not able to pair the bluetooth with the screen. Please follow the steps below:

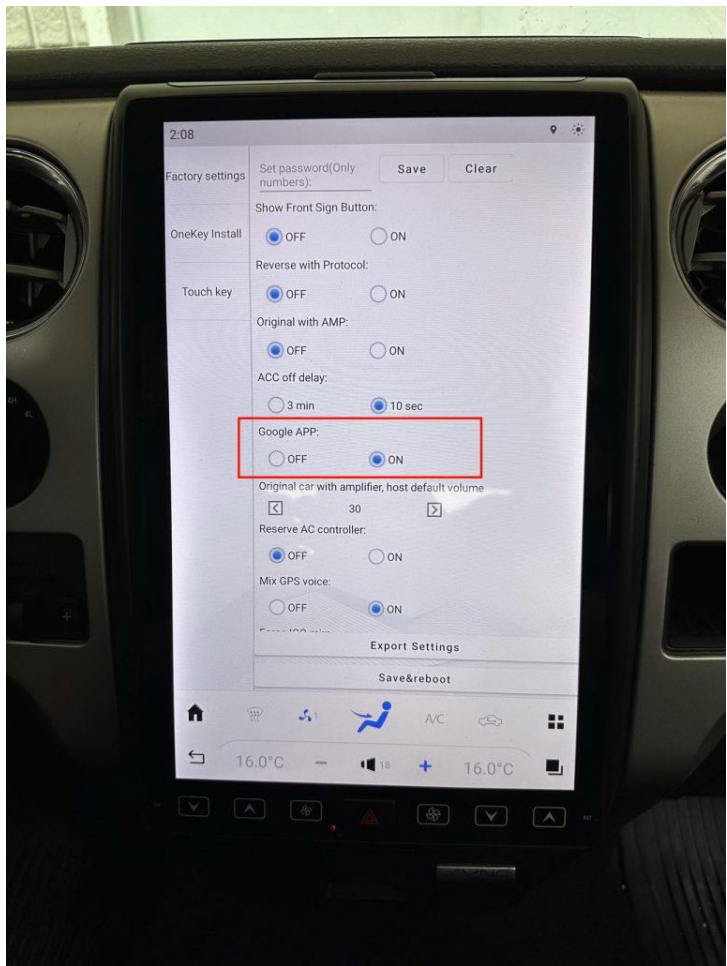
- Go to the Bluetooth menu of your cellphone, search for previous pairing record. Unpair and Delete it.
- Go to the Bluetooth menu of the new system. Click on the Connection icon on the Bluetooth menu.
- Turn off the Bluetooth of the cellphone and turn it back on again.
- Search the Bluetooth name of the new system again on the phone.



[Google App]

After the installation, if you can't use the Google App. Please make sure you have connected to a functioning network and follow the steps below:

1. Connect to the WiFi hotspot of the cellphone
2. Open the internet browser and confirm the internet is working.
3. Make sure the time zone is synchronized to your area.
4. Go to Setting – System – Factory Setting (Password: 7890)
5. Look up a setting “Google App”. Make sure it's turned on
6. Click “Save & Reboot”



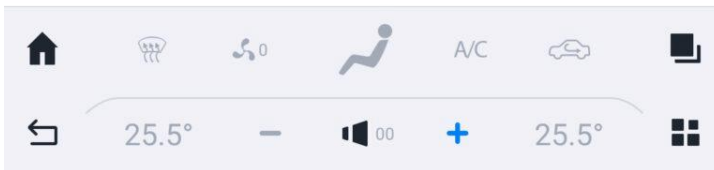
03:59 PM



Google Play

Sign in to find the latest Android apps, games, movies, music, & more

Sign in



[4G LTE]

After inserting the SIM card, please use a small pin (such as a SIM card tray ejector tool) to gently press the RESET button. The reset button is usually located at the front of the unit and labeled as “**RST**”. Once pressed, the system will restart and recognize the SIM card automatically.

Important: Do **not** use any object larger than the ejector pin, as it may crack the screen.

For better compatibility, we recommend using a data-only medium (micro) size SIM card, as cellular voice plans are not supported, only data plans will work. You may also need to configure the APN (Access Point Name) settings depending on your network provider. We have tested the unit with AT&T, Mint Mobile and T-Mobile, and both work well. Please also ensure that the 4G LTE antenna is positioned in a location with strong signal reception. If you experience weak signal or connection issues, try relocating the antenna to another spot with better coverage.

For Mint Mobile: All you need to do is to insert the SIM card with the unit power off and then power the unit back on and after a few seconds it recognizes the SIM card. Mint Mobile will ask you to download their app online whether you use your smartphone or your laptop and so you can activate the card online.

IMEI Number:

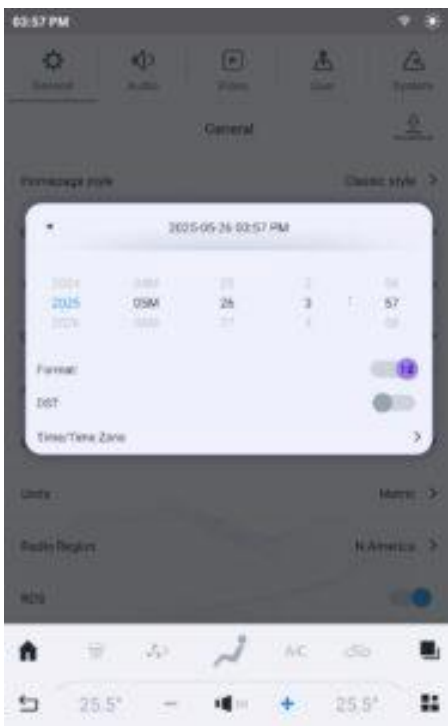
To locate the IMEI number, please follow these steps:

Go to Settings > Android Settings > Mobile Network, and you will find the IMEI number listed there.

In order to access the internet, please make sure the system time is synchronized with your local time.

1. Go to Setting – System – Android – Time Setting
2. Turn ON Network Time
3. Turn OFF auto time zone and manually adjust the time zone according to your area
4. Make sure the time on the upper left corner of the screen matches with your area's local time.
5. Once the time matches with the local time, WiFi should work.
6. To check whether the internet connection is working, you can simply open YouTube in the internet browser. If it loads, the connection is successful.

Secondly, you can go to Setting from the main Android interface, select Time and manually change the date and time under this menu.



[Weather]

How to change the Weather Location on the main interface:

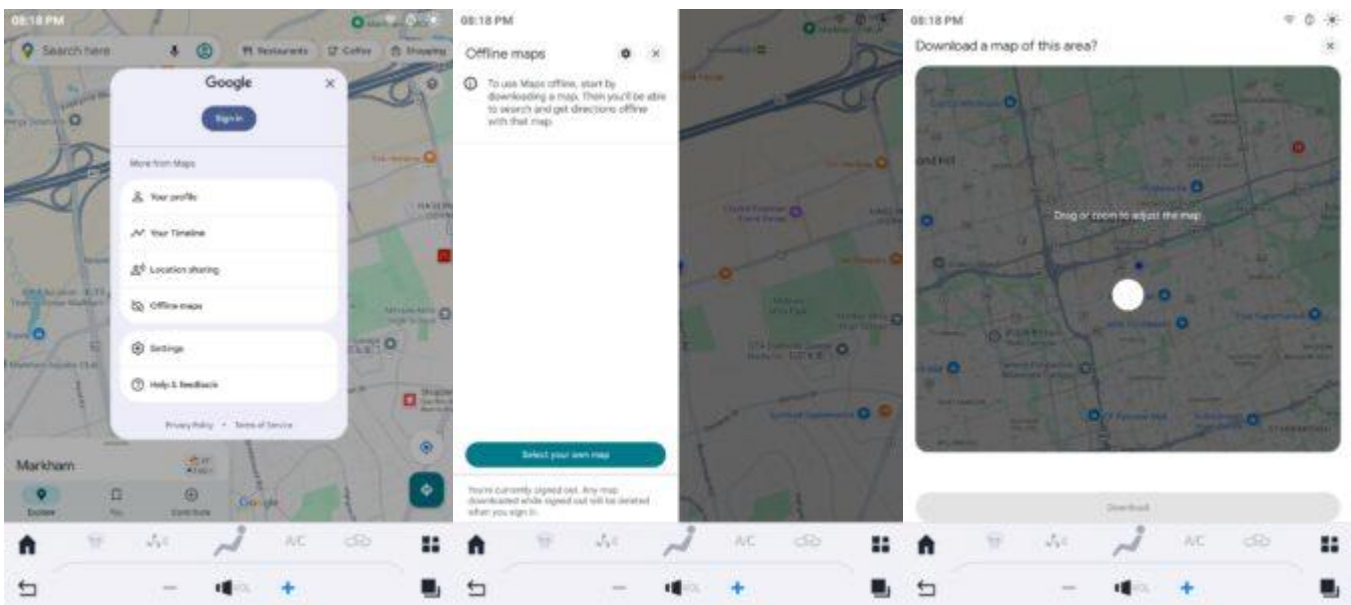


[Offline Navigation]

Using Google Maps for Offline Navigation. To use navigation without an internet connection, follow these steps:

1. Connect to WiFi – Ensure the device is connected to a stable WiFi network.
2. Update Google Maps – Open the Google Play Store, search for Google Maps, and update it to the latest version.
3. Download Offline Maps
4. Open the Google Maps app.
5. Tap your profile icon in the top right corner.
6. Select Offline maps > Select your own map.
7. Zoom and position the area you want to download.
8. Tap Download.

Once the map is downloaded, you'll be able to use navigation and search for locations in that area without an internet connection.



[Apple CarPlay / Android Auto Activation]

If the system asks for activation after a software reset, please follow these steps:

1. Connect to Wi-Fi and ensure the internet is working properly.
2. Verify that the **time zone** and **system time** are correctly set for your location.
3. Open the **Chrome browser** and try to play a video on **YouTube** to confirm that the **internet connection is working properly**.
4. Close the **CarAuto** app completely.

5. Reopen the **CarAuto** app.

The system should activate automatically as long as the internet connection is active.

[Apple CarPlay]

How to switch between Apple CarPlay and WiFi Hotspot:



How to connect / reset wireless Apple CarPlay:

Note: If you have any difficulties on connecting the CarPlay on the new system, please make sure you have **DISCONNECTED** the cellphone from the original Bluetooth first. You can go to the Bluetooth menu of the cellphone to confirm it. If you see any connections with the original system, please disconnect and unpair it before try to connect the CarPlay from the New Android Screen.

Wireless Apple CarPlay only use Bluetooth to pair at the beginning, once it got paired, it will automatically switch to the WiFi channel at the background. While using Wireless Apple CarPlay, you can not turn on WiFi hotspot at the same time. If you would like to turn on the WiFi Hotspot to use some of the applications (such as YouTube..etc), you will need to turn off the CarPlay first. The procedure to turn off the CarPlay shows as below:



CarPlay Reset Procedure:

Important Notes Before Pairing:

- **Disconnect the Original Bluetooth:** Ensure your phone is **disconnected and unpaired** from the vehicle's **original Bluetooth system**. Your phone can only connect to **one Bluetooth system** at a time.
- **Avoid USB Connection to the Original System:** Do **not connect your phone via USB** to the original car system, as it may cause conflicts.
- **Charging Recommendation:** If you need to charge your phone, use a **12V cigarette lighter charger** instead of the USB port. It provides faster charging (12V vs. 5V).
- **Wi-Fi Hotspot Limitation:** Wireless Apple CarPlay or Android Auto cannot be used while your phone's Wi-Fi hotspot is active.

How to Reset & Reconnect CarPlay/Android Auto

Part 1: Clear Previous Connection Data from Phone

1. Erase CarPlay Settings (iPhone only):

- Go to: Settings > General > CarPlay
- Delete all previously connected vehicles.

2. Remove Bluetooth Pairing:

- Go to your phone's Bluetooth settings.
- Locate the Tesla screen's Bluetooth name.
- Select “Forget” or “Unpair”.

3. Restart Your Phone.

Part 2: Clear Previous Connections on Tesla Screen

1. Open the Bluetooth menu on the Tesla screen.
2. Tap the **connection icon** (upper-right corner).
3. **Delete all previously paired phones** from the list.
4. **Restart the Tesla unit:**
 - Use a pin to press the “RST” reset button, usually located at the front of the unit.

How to Connect Wireless Apple CarPlay:

1. On the Tesla Screen:

- Go to the Bluetooth menu.
- Tap the **connection icon**.

2. On Your Phone:

- Go to Bluetooth settings.
- Pair your phone with the new Tesla system.

Note: CarPlay uses Bluetooth only for initial pairing. After pairing, it switches to Wi-Fi automatically in the background. It's normal for Bluetooth to show “Disconnected”—do not reconnect it manually.

3. Once paired, the screen should automatically launch **CarAuto** (CarPlay/Android Auto). If it doesn't, open the **CarAuto app** manually from the app list.

4. On your iPhone, **accept the prompt** to authorize CarPlay connection.

Once setup is complete, you're ready to enjoy Wireless CarPlay/Android Auto!

Guidelines for Using CarPlay/Android Auto:

Connection Requirements

- **Wireless CarPlay/Android Auto uses your phone's WiFi channel** to operate.
- While connected, your phone's WiFi is occupied by CarPlay/Android Auto.
- **Do not manually connect your phone's WiFi hotspot to the Tesla system** while CarPlay/Android Auto is active — this may cause conflicts or connectivity issues.

Automatic Switching Behavior

- After pairing is complete, the system will **automatically switch from Bluetooth to WiFi** to run CarPlay/Android Auto.
- Once connected via WiFi, **Bluetooth will show as “Disconnected”** — this is normal.
- **Do not manually reconnect Bluetooth** after CarPlay/Android Auto is running.

How to Disconnect CarPlay and Use WiFi Hotspot Instead

If you prefer to use your phone's WiFi hotspot for internet on the system rather than CarPlay/Android Auto, follow these steps:

Step 1: Disable CarPlay on iPhone

- Go to: Settings > General > CarPlay
- Find the Tesla system and **turn off CarPlay** (do not delete the connection).

Step 2: Enable Phone's WiFi Hotspot

- Turn on the **WiFi hotspot** feature on your phone.

Step 3: Connect the System to Hotspot

- On the Tesla system, open **WiFi settings**.
- Connect to your phone's hotspot.
- The system may automatically open the **CarAuto app** and disable WiFi again.

If this happens:

- **Close the CarAuto app.**
- **Manually re-enable WiFi** on the Tesla system.

Note: The system will automatically try to launch CarPlay/Android Auto anytime your phone connects via Bluetooth.

To Keep the System in Hotspot Mode

- Open the **CarAuto app settings** on the Tesla screen.
- **Disable the “Auto Connect” option** to prevent automatic switching back to CarPlay/Android Auto.

[Android Auto]

How to connect / reset wireless Android Auto:



How to Prevent CarPlay from Auto-Launching:

To keep the Tesla system connected to your phone's WiFi hotspot instead of automatically launching CarPlay/Android Auto:

1. **Disconnect your phone from the Tesla system** (turn off Bluetooth or forget the connection temporarily).
2. On the Tesla screen, **open the CarAuto app**.
3. Tap the **Settings icon** located at the bottom-right corner.
4. In the Settings menu, **disable the "Auto Connect" option**.

Result: CarPlay/Android Auto will no longer launch automatically when your phone connects. This allows the system to remain in WiFi Hotspot mode for internet access.

[How to Install Aftermarket REVERSE Camera:]

In order to install the backup camera to the system, basically you only need to run the cable from the tailgate to the front unit underneath the vehicle. (Caution: At the back end of the camera wire, there are two wires RED and BLACK, please cut it and tape it. Please make sure both red wire and black wire do not touch each other and cause short circuit. Otherwise, it will burn the camera). Please test the whole camera system and all the wire first before run the wire underneath the truck. Make sure the camera is working before run any camera wire.

The new system provides camera power through the wire labeled "CCD-12," and the camera signal should be connected to the RCA connector labeled "CCD-V." You only need to connect the CCD-12 power wire from the screen to the camera (typically the red power wire on the camera side). There is no need to connect any power from the reverse light at the rear of the vehicle. Instead, you should cut and properly tape off the "RED" and "BLACK" wires at the back end of the camera cable, ensuring they do not come into contact with each other, as any exposed metal could cause a short circuit and potentially damage the camera. Additionally, please go to the reverse settings and change the reversing format to "CVBS." For more detailed instructions, you may continue reading below.

For the wiring instruction, you can also take a look of the video instruction below for a reference.

Below is a video tutorial to show you an introduction of the wiring harness. It may not be exactly the same as your model but it will be similar to all the models.:





Software Settings:

If you are installing the **aftermarket camera**, you will also need to go to Settings – System – Factory Setting (7890 / 666888) – Reversing Format – Change it to “**CVBS**“, then click “Save and Reboot”.

If it's the **OEM camera**, you will also need to go to Settings – System – Factory Setting (7890 / 666888) – Reversing Format – Change it to “**VGA**“, then click “Save and Reboot”.

Trouble Shooting:

- 1) Please prepare a voltage tester. Make sure the wire CCD-12 provide 12V power.
- 2) Please make sure the RED wire at the back end of the camera wire (The end that connect to the reverse camera, it comes with RED and BLACK wire.) Since the system provides 12V power from the front to the camera , there should also be 12V power on the RED wire of the camera side.
- 3) Please make sure the Reversing Camera setting is set to “CVBS”.
- 4) Please make sure the camera wire is not damage.

[How to Install Aftermarket FRONT Camera:]

In order to install the front camera, currently we do not have instruction for it.

- 1) You can simply run the wire from the front to the screen, the camera signal can be connected to AUX-V, and
- 2) In order to provide the power to the camera, you can get the power from any of the accessory wire (+12V) in the vehicle. Please don't get the power directly from the unit, you can get the power from the fuse box of the vehicle.
- 3) In order to activate the front camera, you can click on the icon (AUX) from the app list.

Note: The power out put from the unit (CCD-12 / CCD-6) are only for the reverse camera power. The power can only be activated when the car is in reverse, therefore it can't be used for the front camera.

Camera Image & Parking Guideline Settings

Note: In order to switch the left and right mirror image of the camera, you can connect both “WHITE” and “GREEN” wire at the back end of the camera. The white one control the guidance line and the green one control the mirror image of the camera.

At the back of the front camera, you will find two small control wires: WHITE and GREEN. These wires are used to adjust the camera display settings.

WHITE wire – Parking Guidance Lines

This wire controls the on-screen parking guidelines (the colored lines that help estimate distance when parking). Before connecting or disconnecting these wires, please **disconnect the power from the camera first**. After adjusting the wires as needed, you can **reconnect the power to the camera** to apply the change

Connected (Disable Guideline) → Parking guideline lines will appear on the screen.

Disconnected (Enable Guideline) → Parking guideline lines will be disabled.

GREEN wire – Mirror Image (Left/Right Orientation)

This wire controls whether the camera image is displayed in mirror mode or normal mode. Before connecting or disconnecting these wires, please **disconnect the power from the camera first**. After adjusting the wires as needed, you can **reconnect the power to the camera** to apply the change.

Connected (For Front Camera) → The image will be displayed in mirror mode (left and right are reversed).

Disconnected (For Reverse Camera) → The image will be displayed in normal orientation.

Important:

For a front camera, it is usually recommended to enable the mirror mode so the image displays in the correct orientation.

These wires are located near the camera cable at the back end of the camera, and they can simply be connected together or left disconnected to change the settings.

How to install offline Navigation software and APK files:



How to update MCU / Firmware for the system | Android 13 (If applicable):



How to update System Software for the system | Android 13 (If applicable):



For further technical inquiries, please refer to our technical support page for more details: [HERE](#)